

10. (a) Benzene reacts with bromine in the presence of a catalyst.
Give the mechanism for this reaction and explain how the Br — Br bond becomes polarised during the reaction.
Suggest why, in the absence of this catalyst, there is very little reaction between benzene and bromine under normal conditions. [6 QER]

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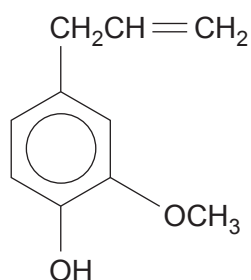
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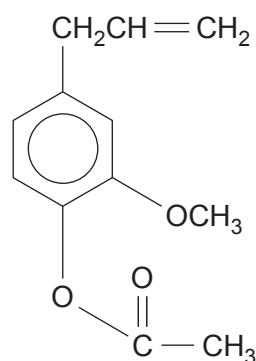
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- (b) Eugenol is the main constituent of clove oil, together with smaller quantities of eugenyl ethanoate.



Eugenol



Eugenyl ethanoate

- (i) Describe what is seen if a few drops of iron(III) chloride solution are added to a solution of eugenol. [1]

- (ii) State the colour change that occurs if a few drops of aqueous bromine are added to a solution of eugenol. [1]

- (iii) If an excess of bromine is added to eugenol, a new compound is formed that contains the following percentages by mass of each element.

C 24.9% H 2.1% O 6.6%, the remainder being the % of bromine

Use this information to calculate the empirical and molecular formulae of this brominated compound and suggest a possible structure for it. [5]



- (iv) Explain how adding aqueous sodium hydroxide at room temperature to a solution of eugenol and eugenyl ethanoate dissolved in trichloromethane, enables the two compounds to be separated.
Trichloromethane and water are immiscible. [2]

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